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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– 24SDCS03

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TITLE OF THE PROJECT		
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Abstract:

Manufacturing industries increasingly rely on automated and intelligent systems to maintain product quality, reduce production waste, and minimize downtime. However, conventional manufacturing systems often detect defects only after they have occurred, resulting in material waste, rework, production delays, and increased operational costs. Digital Twin technology provides a virtual representation of physical manufacturing processes and enables real-time monitoring, while Artificial Intelligence (AI) techniques can support defect prediction and quality analysis. Despite these advancements, existing approaches generally lack an integrated mechanism for real-time hidden defect prediction, root-cause analysis, corrective-action evaluation, and autonomous decision-making. This highlights the need for an intelligent and adaptive system that can continuously monitor manufacturing processes and proactively respond to emerging defects.

To address this need, the proposed research introduces **AgentTwinDefect AI**, an agentic Digital Twin platform designed for real-time hidden defect prediction in smart manufacturing. The system integrates Digital Twin technology, machine learning, sensor-based process monitoring, and AI agents to provide an end-to-end closed-loop quality intelligence framework. The platform collects and

processes manufacturing parameters such as temperature, vibration, speed, and pressure to continuously monitor the physical production process and its virtual representation.

The proposed approach first uses **machine learning models** to identify potential hidden or emerging defects from real-time manufacturing data. Following defect prediction, an **AI-based diagnostic agent** analyzes the available process information to determine the probable root cause of the detected or predicted defect. The Digital Twin then enables the system to **simulate possible corrective actions** virtually, allowing different process adjustments to be evaluated without directly affecting the physical manufacturing system. A decision agent subsequently analyzes the predicted outcomes and recommends the most suitable corrective action.

The proposed framework follows an **agentic closed-loop quality intelligence** mechanism: **Monitor → Predict → Diagnose → Simulate → Recommend → Adapt**. Unlike conventional Digital Twin systems that primarily focus on monitoring and prediction, AgentTwinDefect AI aims to integrate autonomous reasoning, root-cause identification, virtual corrective-action testing, and decision recommendation within a unified platform. The system also incorporates a real-time dashboard to provide monitoring, alerts, predictions, and explanations to support manufacturing operators and decision-makers.

Collectively, AgentTwinDefect AI aims to overcome limitations identified in existing Digital Twin and Agentic AI research, particularly in real-time integration, adaptability, data quality, and end-to-end defect management. By combining predictive intelligence with autonomous agent-based reasoning and Digital Twin simulation, the proposed framework has the potential to enable proactive quality management, reduce defects and production losses, and support faster and more informed corrective decisions in smart manufacturing.

Project Architecture/Project Overview:

